1.9 Copying an Angle

Lesson Introduction

Benchmark	MA.912.GR.5.1 Construct a copy of a segment or an angle.	Learning Targets	• I can construct a copy of an angle. • I can use the angle addition postulate.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What are the steps to construct the copy of an angle? • How can I use the angle addition postulate to determine the value of unknown variables and angle measures?
Lesson Narrative	Students work together to determine the order of the steps needed to construct the copy of an angle. After deciding on an order, they will test the steps, make adjustments as needed, and test again.

Component Summary	1.9.1 Warm-Up (~5 min.)	Bell Work using the segment addition postulate *(SE p. 59)*	1.9.3 Guided Instruction (10 min.)	Angle addition postulate *(SE p. 61)*
	1.9.2 Activity (20 min.)	Determining the steps to construct the copy of an angle *(SE p. 59)*	1.9.4 Try It (10 min.)	Constructing the copy of an angle *(SE p. 62)*
	1.9.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 63)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Angle Addition Postulate <u>Additional Terms:</u> N/A	• Compass • Protractor • Patty Paper • (Optional) Chart Paper	• Make copies of the steps and cut into strips for students to use when determining the order of the steps for constructing the copy of an angle in 1.9.2 Activity.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may think that the length of the rays determines whether an angle is a copy of another.</td><td>• If a document camera or Smart Board are not available, consider making an oversized copy of the steps for constructing a copy of an angle. These steps can be used when providing the final answer and discussion of the step order at the end of 1.9.2 Activity.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Students should work in pairs to order the steps in 1.9.2 Activity. After thinking about the order individually, students can share their progress with their partner. Partner pairs then find another partner pair to share their thinking with. Providing this structure gives students an opportunity to feel confident about their order before testing if it is correct.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students are building perseverance by testing and modifying their steps. Students should be encouraged to stay engaged and maintain a positive mindset, helping and supporting each other through 1.9.2 Activity. Using logic to order the steps is developing skills and a foundation for writing proofs.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 1.9.2 Activity, consider providing students with the correct placement of more of the steps in the beginning so that they have fewer steps to order. For example, provide the correct placement of steps 1, 2, 3, 6, and 9.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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1.9.1 Warm-Up: Bell Work

Component Overview: *Students will use the segment addition postulate to solve for an unknown value and the length of a segment.*

1.9.1 Warm-Up: Bell Work

1. Points C, H, and W are collinear. Point W is between C and H. $CW = 4x - 2$, $HW = 5x - 19$, and $CH = 7x + 5$.

   a. Find the value of x.
   $x = 13$

   b. What is the length of $\overline{HW}$?
   $HW = 46$

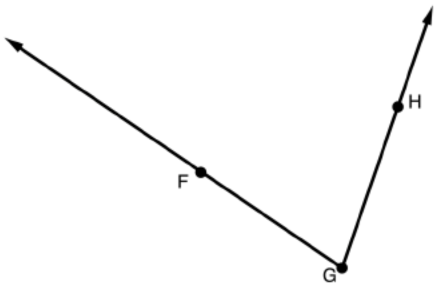
Use student understanding to inform your instruction and to determine where small group instruction may be helpful during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

1.9.2 Activity: Copying an Angle

Component Overview: *Students collaborate on ordering the steps for constructing a copy an angle. After sharing their order with others, the students will follow the steps to copy an angle. Make sure students have adequate time to attempt, fail, and adjust the steps if needed.*

1.9.2 Activity: Copying an Angle

1. $\angle FGH$ is shown.



   a. Discuss with your partner how to copy $\angle FGH$ using only a compass.

   b. Work with your partner to order the nine steps for copying $\angle FGH$. The first and last step are in the correct position.

1	Draw a point and label it X. Then, draw a ray starting at point X in any direction. Label the ray $\overrightarrow{XZ}$.
7	Set the compass' point on point A and adjust the compass' width to point B.
8	Without changing the compass' width, place the compass' point on Y and draw a similar arc to create point W where the two arcs meet.
2	Set the compass' point on point G and adjust the compass' width to any width.
6	Without changing the compass' width, place the compass' point on X and draw a similar arc. Label the point where the arc and $\overrightarrow{XZ}$ intersect point Y.
4 or 5	Label the point where the arc crosses $\overrightarrow{GF}$ as point A.
4 or 5	Label the point where the arc crosses $\overrightarrow{GH}$ as point B.
3	Draw an arc across both rays of $\angle FGH$.
9	Draw $\overrightarrow{XW}$ using a straightedge.

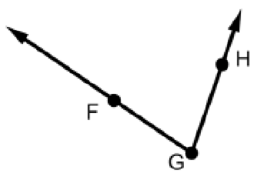
The intention of this activity is to have students collaborate to order the steps for constructing the copy of an angle. Consider making copies of the provided steps and cut them into strips. Have the students put the strips in the order they think will provide the steps for constructing the copy of an angle. Student partners should compare their ordering with other partner groups. To verify if their ordering is correct, students should follow the steps to construct the copy of an angle. If the steps are not in the correct order, students can adjust to correct themselves and verify again. Once the entire activity is completed, students can number the steps in the correct order in the table below for future reference.

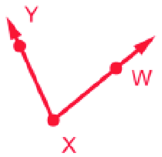
Students learn more by making mistakes. Provide students with adequate time to order the steps, test the steps, and adjust the steps to try again.

MA.K12.MTR.1.1: Participating in Effortful Learning Students are building perseverance by testing and modifying their steps. Students should be encouraged to stay engaged and maintain a positive mindset, helping and supporting each other through the activity. Using logic to order the steps is developing skills and a foundation for writing proofs.

c. Compare your ordering with another group. If necessary, make adjustments to the order.
Answers will vary.

d. Use your steps to draw $\angle WXY$, a copy of $\angle FGH$. If necessary, make adjustments to the order of your steps.





To review the correct ordering, select a group to share their ordering with the class. This can be done with a document camera, a Smart Board, or by making large copies of the steps that can be placed in order on the board. If a classmate doesn't agree with the ordering, the classmate should question the presenters and provide reasoning for why the order is incorrect. This continues until the class agrees upon the correct order.

Consider allowing students to edit the steps. Some students may consider the process easier if some steps are combined. Help students understand that memorizing the steps is not the point. This can be related to common tasks students do that require steps, but they do not have the steps memorized. Tying one's shoes is such a task. Students know the process but do not need to have the steps memorized. Alternately, they could likely figure out if steps are correct based on their experience.

1.9.3 Guided Instruction: Angle Addition Postulate

Component Overview: *Students discover the angle addition postulate.*

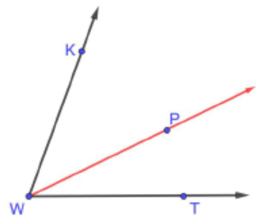
1.9.3 Guided Instruction: Angle Addition Postulate

Angle Addition Postulate

If point X is in the interior of $\angle AMB$, then $m\angle AMX + m\angle XMB = m\angle AMB$.

The Angle Addition Postulate typically makes sense to students. Use 1.9.3 Guided Instruction as an opportunity to remind students the difference between notations such as $\angle ABC$ and $m\angle ABC$.

1. Point P lies in the interior of $\angle KWT$ as shown.



   a. Draw $\overrightarrow{WP}$.
   Shown on diagram.

   b. Use a protractor to measure each angle in degrees.
   $m\angle KWT = 70°$ $m\angle KWP = 44°$ $m\angle PWT = 26°$

   c. Discuss with your partner why $m\angle KWP + m\angle PWT = m\angle KWT$ should be true. Summarize your discussion.
   The two smaller angles should add up to the measurement of the large angle.

   d. If $m\angle KWT = 70°$, $m\angle KWP = (5x + 5)°$ and $m\angle PWT = (4x - 7)°$, determine the value of x.
   $x = 8$

1.9.4 Try It: Copying an Angle

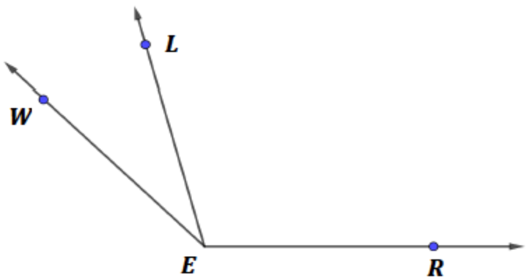
Component Overview: *Students construct a copy of a given angle.*

1.9.4 Try It: Copying an Angle

1. $\angle KLM$ is shown.

Draw $\angle RST$, a copy of $\angle KLM$.

2. Three angles are shown, $\angle WEL$, $\angle LER$, and $\angle WER$.



If $m\angle LER = (14.35x - 9.8)°$, $m\angle WEL = (2.5x + 10)°$, and $m\angle WER = 135°$, what is $m\angle WEL$?

$m\angle WEL = 30°$

Monitor students to make sure the appropriate construction marks are placed to ensure the correct construction. Also, monitor that students are labeling the points on the copy, as this is a necessary part of the process. Students can use patty paper or a protractor to assess the accuracy of the copy.

1.9.5 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

1.9.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about copying an angle.

   Answers will vary.

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.